

Information Retrieval

Evaluation

This lecture

❖ How do we know if our results are any good?

– Evaluating a search engine

- Benchmarks
- Precision and recall

❖ Results presentation:

– Making our good results usable to a user

Measures for a search engine

- ❖ How fast does it **index**
 - Number of documents/hour
 - (Average document size)
- ❖ How fast does it **search**
 - Latency as a function of index size
- ❖ Expressiveness of query language
 - Ability to express complex information needs
 - Speed on complex queries
- ❖ Nice UI
- ❖ Other features
 - Spelling, similarity, query expansion, semantics

Efficiency Metrics

Metric name	Description
Elapsed indexing time	Measures the amount of time necessary to build a document index on a particular system.
Indexing processor time	Measures the CPU seconds used in building a document index. This is similar to elapsed time, but does not count time waiting for I/O or speed gains from parallelism.
Query throughput	Number of queries processed per second.
Query latency	The amount of time a user must wait after issuing a query before receiving a response, measured in milliseconds. This can be measured using the mean, but is often more instructive when used with the median or a percentile bound.
Indexing temporary space	Amount of temporary disk space used while creating an index.
Index size	Amount of storage necessary to store the index files.

Measures for a search engine

❖ Many of these criteria are measurable

❖ But the key measure is user happiness

- What is this?
- Speed of response/size of index are factors
- But blindingly fast, useless answers won't make a user happy

❖ Need a way of quantifying user happiness

Metric name	Description
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Measuring user happiness

- ❖ Who is the user we are trying to make happy?
 - Depends on the setting
- ❖ Web engine:
 - User finds what they want and return to the engine
 - Can measure rate of return users
 - User completes their task – search as a means, not end
- ❖ eCommerce site:
 - User finds what they want and buy
 - Is it the end-user, or the eCommerce site, whose happiness we measure?
 - Measure time to purchase, or fraction of searchers who become buyers?
- ❖ Institutional: Care about “user productivity”
 - How much time do my users save when looking for information?
 - Many other criteria having to do with breadth of access, secure access, etc.

Happiness: elusive to measure

- ❖ Most common proxy: **relevance of search results**
- ❖ Relevance measurement requires 3 elements:
 - A benchmark document collection
 - A benchmark suite of queries
 - A usually binary assessment of either Relevant or Nonrelevant for each query and each document

Evaluating an IR system

- ❖ Note: the **information need** is translated into a **query**
- ❖ Relevance is assessed relative to the **information need** *not* the **query**
- ❖ E.g., Information need: *I'm looking for information on whether drinking red wine is more effective at reducing your risk of heart attacks than white wine.*
- ❖ Query: **wine red white heart attack effective**
- ❖ We evaluate whether the doc addresses the information need, not whether it has these words

Standard relevance benchmarks

❖ The Text Retrieval Conference (**TREC**)

- co-sponsored by the National Institute of Standards and Technology (NIST) has run a large IR test bed for many years
- <https://trec.nist.gov/>

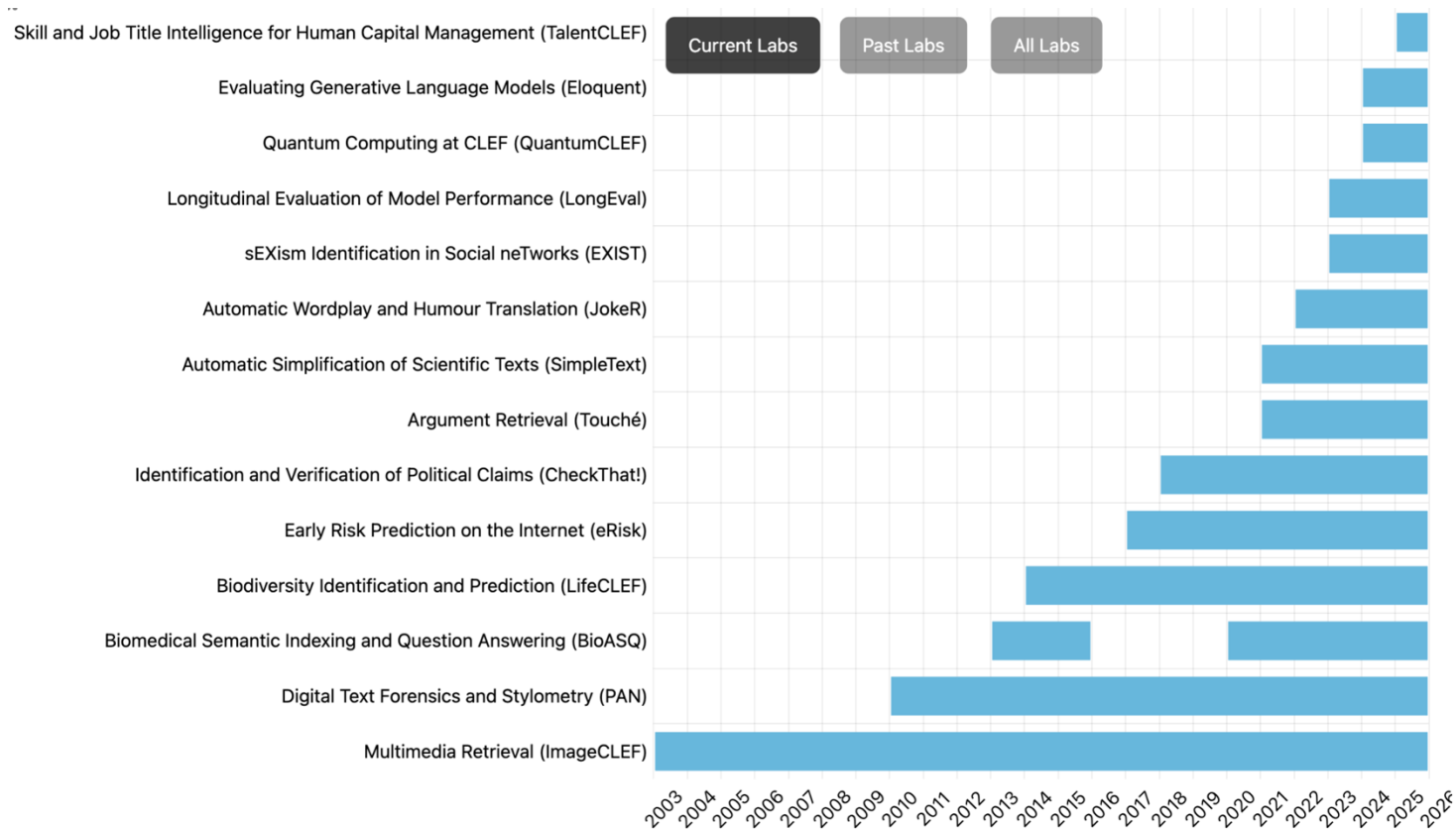
❖ Conference and Labs of the Evaluation Forum (**CLEF**)

- Promotes research, innovation, and development of information access systems with an emphasis on multilingual and multimodal information
- <https://www.clef-initiative.eu/>

❖ Human experts mark, for each query and each doc, Relevant or Nonrelevant

- or at least for a subset of docs that some system returned for that query

Standard relevance benchmarks – CLEF



Precision and Recall

❖ Precision

- fraction of retrieved docs that are relevant = $P(\text{relevant}|\text{retrieved})$
- Precision is used when probability that a positive result is correct is important

❖ Recall

- fraction of relevant docs that are retrieved = $P(\text{retrieved}|\text{relevant})$

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

$$\text{Precision } \mathbf{P} = \text{tp}/(\text{tp} + \text{fp})$$

$$\text{Recall } \mathbf{R} = \text{tp}/(\text{tp} + \text{fn})$$

Classification Errors

❖ False Positive (Type I error)

- a non-relevant document is retrieved

$$\text{Fallout} = \text{FPR} = \text{fp}/(\text{tn}+\text{fp})$$

❖ False Negative (Type II error)

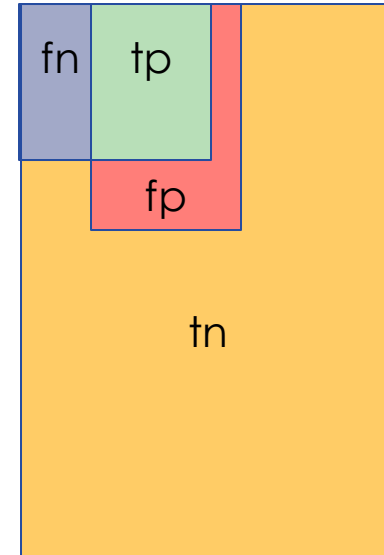
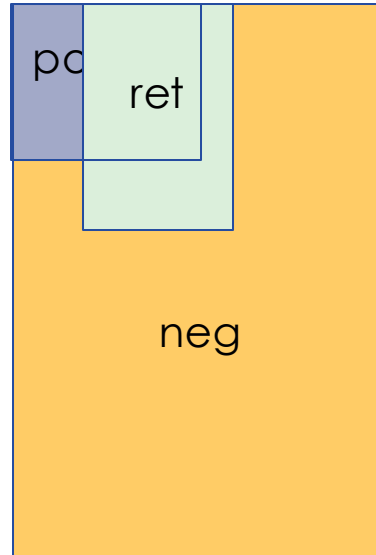
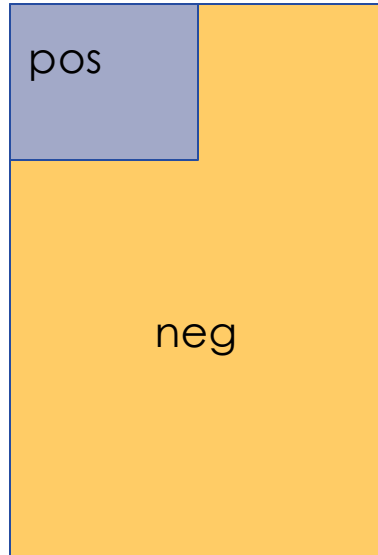
- a relevant document is not retrieved

$$\text{fn}/(\text{tp}+\text{fn})$$
$$=$$

1- Recall

(= FNR = 1-TPR = 1-Sensitivity)

Precision and Recall



$$\text{Precision } P = \frac{tp}{tp + fp}$$

$$\text{Recall } R = \frac{tp}{tp + fn}$$

Binary classification measures

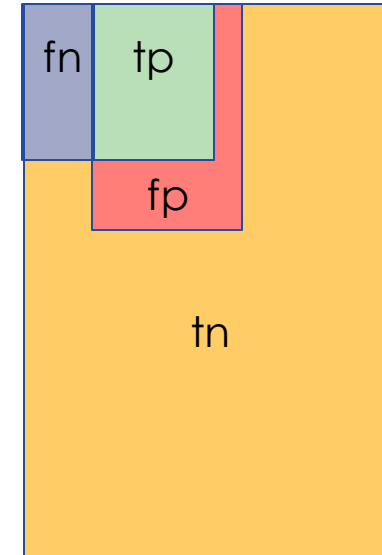
		Condition (as determined by "Gold standard")		
		Condition Positive	Condition Negative	
Test Outcome	Test Outcome Positive	True Positive	False Positive (Type I error)	Positive predictive value = $\frac{\Sigma \text{ True Positive}}{\Sigma \text{ Test Outcome Positive}}$
	Test Outcome Negative	False Negative (Type II error)	True Negative	Negative predictive value = $\frac{\Sigma \text{ True Negative}}{\Sigma \text{ Test Outcome Negative}}$
		Sensitivity = $\frac{\Sigma \text{ True Positive}}{\Sigma \text{ Condition Positive}}$	Specificity = $\frac{\Sigma \text{ True Negative}}{\Sigma \text{ Condition Negative}}$	
		Recall		Precision

Accuracy

- ❖ Given a query, an engine classifies each doc as “Relevant” or “Nonrelevant”
- ❖ The accuracy of an engine: the fraction of these classifications that are correct
 - $(tp + tn) / (tp + fp + fn + tn)$
- ❖ Accuracy is a commonly used evaluation measure in machine learning classification work
- ❖ Why is this not a very useful evaluation measure in IR?

Why not just use accuracy?

- ❖ How to build a highly accurate search engine on a low budget?
- ❖ Typically :
 - $(tp + tn) / (tp + fp + fn + tn)$
 - $tn \gg tp$



Precision/Recall

- ❖ We can get high recall (but low precision) by retrieving all docs for all queries!
- ❖ Recall is a non-decreasing function of the number of docs retrieved
- ❖ In a good system, precision decreases as either the number of docs retrieved or recall increases
 - This is not a theorem, but a result with strong empirical confirmation
- ❖ **Q:** How to adequately combine the Precision and Recall measures?

F Measure

- ❖ The harmonic mean of recall and precision (**F1**)

$$F_1 = \frac{1}{\frac{1}{2}(\frac{1}{R} + \frac{1}{P})} = \frac{2RP}{(R+P)}$$

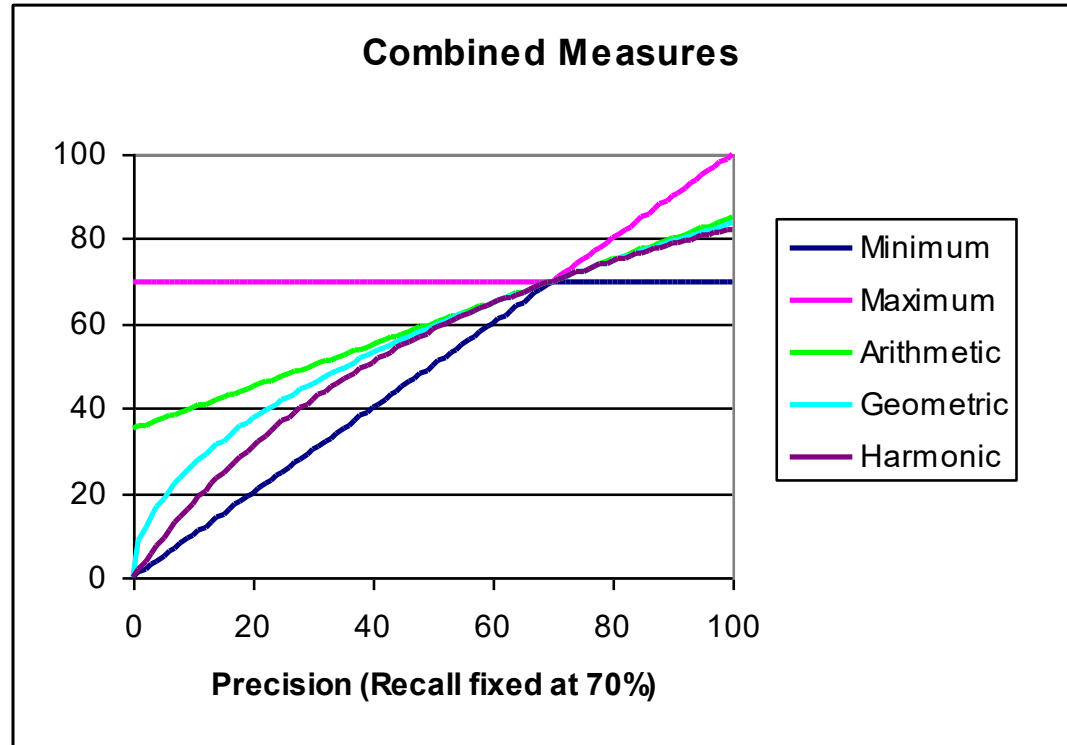
- harmonic mean emphasizes the importance of small values, whereas the arithmetic mean is affected more by outliers that are unusually large

- ❖ More general form

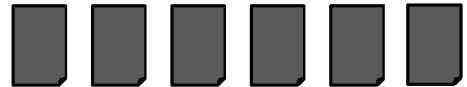
$$F_\beta = (\beta^2 + 1)RP / (R + \beta^2 P)$$

- β is a parameter that determines relative importance of recall and precision

F1 and other averages



Ranking Effectiveness

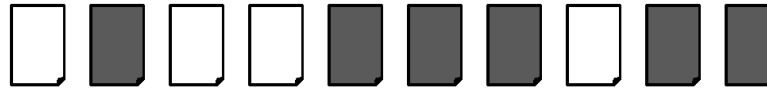
 = the relevant documents

Ranking #1



Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2



Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

Summarizing a Ranking

- ❖ Calculating recall and precision at fixed rank positions
- ❖ Calculating precision at standard recall levels, from 0.0 to 1.0
 - requires *interpolation*
- ❖ Averaging the precision values from the rank positions where a relevant document was retrieved

Average Precision

 = the relevant documents

Ranking #1

										
Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2

										
Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

$$\text{Ranking \#1: } (1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$$

$$\text{Ranking \#2: } (0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$$

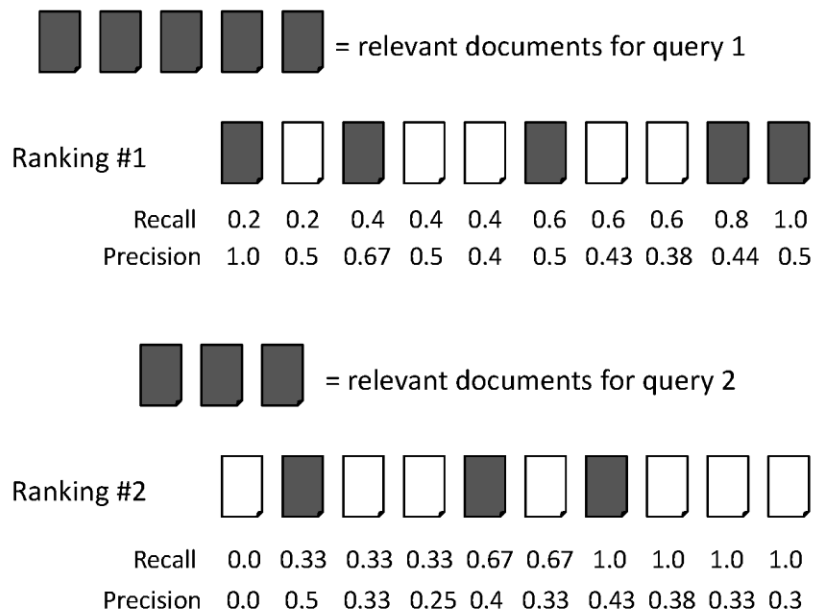
MAP – Averaging Across Queries

❖ Mean Average Precision (MAP)

- summarize rankings from multiple queries by averaging the average precision
- most commonly used measure in research papers
- assumes the user is interested in finding many relevant documents for each query
- requires many relevance judgments in a text collection

❖ Recall-precision graphs are also useful summaries

MAP – Averaging Across Queries



$$\text{average precision query 1} = (1.0 + 0.67 + 0.5 + 0.44 + 0.5)/5 = 0.62$$

$$\text{average precision query 2} = (0.5 + 0.4 + 0.43)/3 = 0.44$$

$$\text{mean average precision} = (0.62 + 0.44)/2 = 0.53$$

Focusing on Top Documents

- ❖ Users tend to look at only the top part of the ranked result list to find relevant documents
- ❖ Some search tasks have only one relevant document
 - e.g., navigational search, question answering
- ❖ Recall not appropriate
 - instead need to measure how well the search engine does at retrieving relevant documents at very high ranks

Focusing on Top Documents

❖ Precision at Rank R

- R typically 5, 10, 20
- easy to compute, average, understand
- not sensitive to rank positions less than R

Ranking #1										
Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6
Ranking #2										
Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

❖ Reciprocal Rank (RR)

- reciprocal of the rank at which the first relevant document is retrieved

$$d_p, d_n, d_p, d_p, d_p \rightarrow (RR = 1)$$

$$d_n, d_p, d_n, d_n, d_p \rightarrow (RR = 0.5)$$

- very sensitive to rank position

MRR

- ❖ Mean Reciprocal Rank (MRR) is the average of the reciprocal ranks over a set of queries

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^Q \frac{1}{\text{rank}_i}.$$

Query	Results	Correct response	Rank	Reciprocal rank
cat	catten, cati, cats	cats	3	1/3
torus	torii, tori , toruses	tori	2	1/2
virus	viruses , virii, viri	viruses	1	1

Discounted Cumulative Gain

- ❖ Popular measure for evaluating web search and related tasks
 - Uses *graded relevance* as a measure of the usefulness, or *gain*, from examining a document
- ❖ Two assumptions:
 - Highly relevant documents are more useful than marginally relevant document
 - the lower the ranked position of a relevant document, the less useful it is for the user, since it is less likely to be examined
- ❖ Gain is accumulated starting at the top of the ranking and may be reduced, or *discounted*, at lower ranks
- ❖ Typical discount is $1/\log(\text{rank})$
 - With base 2, the discount at rank 4 is $1/2$, and at rank 8 it is $1/3$

Discounted Cumulative Gain

- ❖ DCG is the total gain accumulated at a particular rank p :

$$DCG_p = rel_1 + \sum_{i=2}^p \frac{rel_i}{\log_2 i}$$

Example:

- ❖ 10 ranked documents judged on 0-3 relevance scale:

3, 2, 3, 0, 0, 1, 2, 2, 3, 0

- ❖ discounted gain:

3, 2/1, 3/1.59, 0, 0, 1/2.59, 2/2.81, 2/3, 3/3.17, 0

= 3, 2, 1.89, 0, 0, 0.39, 0.71, 0.67, 0.95, 0

- ❖ DCG ($p = [1..10]$):

3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61

Normalized DCG (nDCG)

- ❖ DCG numbers are averaged across a set of queries at specific rank values
 - e.g., DCG at rank 5 is 6.89 and at rank 10 is 9.61
- ❖ DCG values are often *normalized* by comparing the DCG at each rank with the DCG value for the *perfect ranking*
 - makes averaging easier for queries with different numbers of relevant documents

nDCG Example

- ❖ Perfect ranking:

3, 3, 3, 2, 2, 2, 1, 0, 0, 0

- ❖ ideal DCG values:

3, 6, 7.89, 8.89, 9.75, 10.52, 10.88, 10.88, 10.88, 10.88

- ❖ Real DCG values (previous example):

3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61

- ❖ NDCG values (divide actual by ideal):

1, 0.83, 0.87, 0.76, 0.71, 0.69, 0.73, 0.8, 0.88, 0.88

– $NDCG \leq 1$ at any rank position

This lecture

❖ How do we know if our results are any good?

– Evaluating a search engine

- Benchmarks
- Precision and recall

❖ Results presentation:

– Making our good results usable to a user

Result Summaries

- ❖ Having ranked the documents matching a query, we wish to present a results list
- ❖ Most commonly, a list of the document titles plus a short summary, aka “10 blue links”

 Wikipedia
[https://pt.wikipedia.org/wiki/Lu...](https://pt.wikipedia.org/wiki/Lu%C3%AD_Vaz_de_Cam%C3%B5es) · [Translate this page](#) :

Luís de Camões – Wikipédia, a enciclopédia livre

Luís Vaz de Camões (Lisboa^[?], c. 1524 – Lisboa, 10 de junho de 1579 ou 1580) foi um poeta nacional de Portugal, considerado uma das maiores figuras da ...

 Camões - Instituto da Cooperação e da Língua
<https://www.instituto-camoes.pt> · [Translate this page](#) :

Início - Camões - Instituto da Cooperação e da Língua

Camões, Ministério dos Negócios Estrangeiros, Portugal. Promoção da Língua e Cultura Portuguesas.

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 infopedia.pt - Porto Editora
<https://www.infopedia.pt/artigos> · [Translate this page](#) :

Luís de Camões - Infopédia

Poeta português, filho de Simão Vaz de Camões e de Ana de Sá e Macedo, Luís Vaz de Camões terá nascido por volta de 1524/1525, não se sabe exatamente onde, ...

 RTP Ensina
<https://ensina.rtp.pt/Artigos> · [Translate this page](#) :

Luís Vaz de Camões, o poeta da epopeia dos ...

Luís Vaz de Camões (1524?-1580) é autor dos Lusíadas, o poema épico sobre as descobertas portuguesas. A importância da sua obra só foi reconhecida após a ...

Summaries

- ❖ The title is often automatically extracted from document metadata. What about the summaries?
 - This description is crucial.
 - User can identify good/relevant hits based on description.

- ❖ Two basic kinds:
 - A **static summary** of a document is always the same, regardless of the query that hit the doc
 - A **dynamic summary** is a *query-dependent* attempt to explain why the document was retrieved for the query at hand

Static summaries

- ❖ In typical systems, the static summary is a subset of the document
- ❖ Simplest heuristic: the first 50 (or so – this can be varied) words of the document
 - Summary cached at indexing time
- ❖ More sophisticated: extract from each document a set of “key” sentences
 - Simple NLP heuristics to score each sentence
 - Summary is made up of top-scoring sentences

Dynamic summaries

- ❖ Dynamic summaries are tailored in real-time to reflect the most relevant information based on user needs.
- ❖ Present one or more “windows” within the document that contain several of the query terms

camões vasco da gama

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Luís Vaz de **Camões** terá nascido no mesmo ano em que **Vasco da Gama** faleceu. Capitão-mor da expedição que descobriu o caminho marítimo para a Índia entre 1497 e 1499, **Gama** faleceu na Índia, em Cochim, mas foi **Camões** quem lhe deu a fama mundial ao torná-lo no protagonista da sua epopeia, Os Lusíadas.

Curso ONLINE 12
NO CENTENÁRIO DE LUIS VAZ DE CAMÕES E VASCO DA GAMA
Coordenação: João Paulo Oliveira e Costa

 Centro Nacional de Cultura #CNC
<https://www.cnc.pt> > no-centenario-de-luis-vaz-de-camoe... :

No centenário de Luís Vaz de Camões e de Vasco da Gama

Quicklinks

- ❖ For a navigational query such as **camões instituto** user's need is likely satisfied on [instituto-camoes.pt/](https://www.instituto-camoes.pt/)
- ❖ Quicklinks provide navigational cues on that home page



This lecture

- ❖ How do we know if our results are any good?
 - Benchmarks
 - Precision, Recall, F-measure, DCC, nDCC
- ❖ Results presentation:
 - Clear Layout: Organizing results in a visually appealing and easy-to-navigate format.
 - Summarization: Providing concise summaries or snippets of each result to help users quickly identify relevance.
 - Filtering Options: Allowing users to filter results by date, type, or relevance.
 - Ranking: Displaying the most relevant results first, based on the evaluation metrics.
 - User Feedback Mechanism: Implementing features for users to rate or provide feedback on results, helping to refine future searches.